Mathematical Foundations of Political Methodology

Robert Gulotty

University of Chicago

November 4, 2019

Lecture 11 Conditional probability

- Conditional Probability
- Bayes Rule
- Independence

Why study conditional probability?

Most of the relationships we are interested in as policy makers/analysts, are conditional ones. Knowing that an event has occurred might lead us to reassess the probability of another event.

Why study conditional probability?

Most of the relationships we are interested in as policy makers/analysts, are conditional ones. Knowing that an event has occurred might lead us to reassess the probability of another event.

- What is the probability of Iran obtaining nuclear weapons given that it has a uranium enrichment program?

Why study conditional probability?

Most of the relationships we are interested in as policy makers/analysts, are conditional ones. Knowing that an event has occurred might lead us to reassess the probability of another event.

- What is the probability of Iran obtaining nuclear weapons given that it has a uranium enrichment program?
- What is the probability that we will be able to affect climate change if we adopt the climate change treaty?

Why study conditional probability?

Most of the relationships we are interested in as policy makers/analysts, are conditional ones. Knowing that an event has occurred might lead us to reassess the probability of another event.

- What is the probability of Iran obtaining nuclear weapons given that it has a uranium enrichment program?
- What is the probability that we will be able to affect climate change if we adopt the climate change treaty?
- What is the probability of holding liberal political views given certain demographic characteristics?

Definition of Conditional probability

The *conditional probability* of an event A given an event B is given by:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

for all $P(B) \neq 0$

Definition of Conditional probability

The *conditional probability* of an event A given an event B is given by:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

for all $P(B) \neq 0$
We can rearrange this to form the "Multiplication Rule":

$$P(A \cap B) = P(A|B) \times P(B)$$

Examples of Conditional Probability

Example 1:

Examples of Conditional Probability

Example 1:

- $F = \{\text{Myanmar sells oil to China}\}$

Examples of Conditional Probability

Example 1:

- $F = \{\text{Myanmar sells oil to China}\}$
- $E = \{\text{Myanmar discovers oil}\}$

Examples of Conditional Probability

Example 1:

- $F = \{\text{Myanmar sells oil to China}\}$
- $E = \{\text{Myanmar discovers oil}\}$
- If F occurs then E must occur, $P(E|F) = 1$

Examples of Conditional Probability

Example 1:

- $F = \{\text{Myanmar sells oil to China}\}$
- $E = \{\text{Myanmar discovers oil}\}$
- If F occurs then E must occur, $P(E|F) = 1$

Example 2: Security Council voting on Syria strike

Examples of Conditional Probability

Example 1:

- $F = \{$Myanmar sells oil to China$\}$
- $E = \{$Myanmar discovers oil$\}$
- If F occurs then E must occur, $P(E|F) = 1$

Example 2: Security Council voting on Syria strike

- $V = \{$Russia and China veto strike$\}$

Examples of Conditional Probability

Example 1:

- $F = \{$Myanmar sells oil to China$\}$
- $E = \{$Myanmar discovers oil$\}$
- If F occurs then E must occur, $P(E|F) = 1$

Example 2: Security Council voting on Syria strike

- $V = \{$Russia and China veto strike$\}$
- $A = \{$Strike approved by UN$\}$

Examples of Conditional Probability

Example 1:

- $F = \{\text{Myanmar sells oil to China}\}$
- $E = \{\text{Myanmar discovers oil}\}$
- If F occurs then E must occur, $P(E|F) = 1$

Example 2: Security Council voting on Syria strike

- $V = \{\text{Russia and China veto strike}\}$
- $A = \{\text{Strike approved by UN}\}$
- $P(A|V) = \frac{P(A \cap V)}{P(V)} = \frac{P(\emptyset)}{P(V)} = 0$

Axioms of Conditional probability

$P(A|B)$ is a probability, so it follows the axioms.

1. $P(A|B) \geq 0, \quad \forall A$
2. $P(\Omega|B) = 1.$
3. For all mutually exclusive events, $A_1, \ A_2, \ ...A_n$

$$P(A_1 \cup A_2 \cup ... \cup A_n|B) = P(A_1|B) + P(A_2|B) + ... + P(A_n|B)$$

Conditional Probability Example

Basic claim: Foreign military threats (given certain other conditions) produce revolutions.

- Skocpol (1979)
- Let E denote the event of a revolution and
- Let F the presence of a foreign threat

Conditional Probability Example

Basic claim: Foreign military threats (given certain other conditions) produce revolutions.

- Skocpol (1979)
- Let E denote the event of a revolution and
- Let F the presence of a foreign threat

The way to read Skocpol's hypothesis in probability language is:
$P(E|F) \geq P(E|F^c)$

Conditional Probability Example

How might we know whether $P(E|F) > P(E|F^c)$?

Conditional Probability Example

How might we know whether $P(E|F) > P(E|F^c)$?

- $P(E|F) = \frac{P(E \cap F)}{P(F)}$
- $P(E|F^c) = \frac{P(E \cap F^c)}{P(F^c)}$

Conditional Probability Example

How might we know whether $P(E|F) > P(E|F^c)$?

- $P(E|F) = \frac{P(E \cap F)}{P(F)}$
- $P(E|F^c) = \frac{P(E \cap F^c)}{P(F^c)}$
- So we need to know both the rate of Revolutions with foreign threats, the rate of Revolutions without foreign threats, and the probability of a foreign threat.

Revolutions

Skocpol's data:

	Revolution	No Revolution
Foreign Threat	France, Russia, China	
No Threat		England, Prussia, Japan

$P(A|B)$ and $P(B|A)$ are different

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(A|B)$ and $P(B|A)$ are different

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$P(A|B)$ and $P(B|A)$ are different

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Example 1

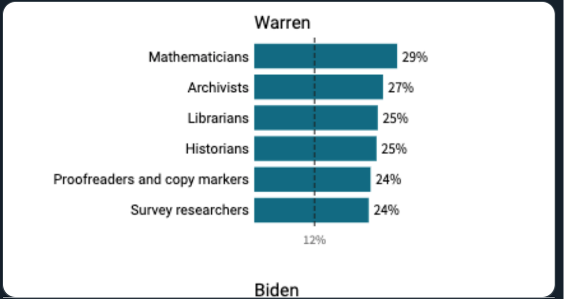
$$P(\text{Tall}|\text{Professional Basketball Player}) = .9$$

$$P(\text{Professional Basketball Player}|\text{Tall}) = .00002$$

Example 2

Jeet Heer @HeerJeet

This list of top jobs among Warren donors seems very on brand.



6:30 PM · Oct 28, 2019 · Twitter Web Client

143 Retweets **871** Likes

Proposition

Multiplication Rule: Suppose $E_1, E_2, \ldots, E_N$ is a sequence of events.

$$P(E_1 \cap E_2 \cap \cdots \cap E_N) = P(E_1)P(E_2|E_1)P(E_3|E_2, E_1) \times \cdots \times P(E_N|E_{N-1}, E_{N-2}, \ldots, E_1)$$

Proposition

Multiplication Rule: Suppose $E_1, E_2, \ldots, E_N$ is a sequence of events.

$$P(E_1 \cap E_2 \cap \cdots \cap E_N) = P(E_1)P(E_2|E_1)P(E_3|E_2, E_1) \times \cdots \times P(E_N|E_{N-1}, E_{N-2}, \ldots, E_1)$$

Proof.

$$\begin{aligned} P(E_1)P(E_2|E_1) &= P(E_1)\frac{P(E_2 \cap E_1)}{P(E_1)} \quad \textbf{Def of CP} \\ &= P(E_1 \cap E_2) \end{aligned}$$

Proposition

Multiplication Rule: Suppose $E_1, E_2, \ldots, E_N$ is a sequence of events.

$$P(E_1 \cap E_2 \cap \cdots \cap E_N) = P(E_1)P(E_2|E_1)P(E_3|E_2, E_1) \times \cdots \times P(E_N|E_{N-1}, E_{N-2}, \ldots, E_1)$$

Proof.

$$\begin{aligned}
P(E_1)P(E_2|E_1) &= P(E_1)\frac{P(E_2 \cap E_1)}{P(E_1)} \quad \textbf{Def of CP} \\
&= P(E_1 \cap E_2) \\
P(E_1 \cap E_2)P(E_3|E_1, E_2) &= P(E_1 \cap E_2)\frac{P(E_3 \cap E_2 \cap E_1)}{P(E_2 \cap E_1)} \\
&= P(E_3 \cap E_2 \cap E_1)
\end{aligned}$$

Proposition

Multiplication Rule: Suppose $E_1, E_2, \ldots, E_N$ is a sequence of events.

$$\begin{aligned} P(E_1 \cap E_2 \cap \cdots \cap E_N) &= \\ P(E_1)P(E_2|E_1)P(E_3|E_2, E_1) \times \cdots \times P(E_N|E_{N-1}, E_{N-2}, \ldots, E_1) \end{aligned}$$

Proof.

$$\begin{aligned} P(E_1)P(E_2|E_1) &= P(E_1)\frac{P(E_2 \cap E_1)}{P(E_1)} \quad \textbf{Def of CP} \\ &= P(E_1 \cap E_2) \\ P(E_1 \cap E_2)P(E_3|E_1, E_2) &= P(E_1 \cap E_2)\frac{P(E_3 \cap E_2 \cap E_1)}{P(E_2 \cap E_1)} \\ &= P(E_3 \cap E_2 \cap E_1) \end{aligned}$$

Repeating for all probabilities proves the proposition

$\square$

Law of Total Probability

For a set of mutually exclusive events, $E_1, E_2, E_3....E_n$, which together form the sample space Ω, the probability of any event A is:

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + ... + P(A|E_n)P(E_n)$$

Law of Total Probability

For a set of mutually exclusive events, $E_1,\ E_2,\ E_3....E_n$, which together form the sample space Ω, the probability of any event A is:

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + ... + P(A|E_n)P(E_n)$$

Proof.

$$P(A) \quad = \quad P(A \cap \Omega)$$

Law of Total Probability

For a set of mutually exclusive events, $E_1, E_2, E_3....E_n$, which together form the sample space Ω, the probability of any event A is:

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + ... + P(A|E_n)P(E_n)$$

Proof.

$$\begin{aligned} P(A) &= P(A \cap \Omega) \\ P(A) &= P(A \cap (E_1 \cup E_2 \cup ... \cup E_n)) \end{aligned}$$

Law of Total Probability

For a set of mutually exclusive events, $E_1,\ E_2,\ E_3....E_n$, which together form the sample space Ω, the probability of any event A is:

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + ... + P(A|E_n)P(E_n)$$

Proof.

$$\begin{aligned} P(A) &= P(A \cap \Omega) \\ P(A) &= P(A \cap (E_1 \cup E_2 \cup ... \cup E_n)) \\ P(A) &= P((A \cap E_1) \cup (A \cap E_2) \cup ... \cup (A \cap E_n)) \end{aligned}$$

Law of Total Probability

For a set of mutually exclusive events, $E_1,\ E_2,\ E_3....E_n$, which together form the sample space Ω, the probability of any event A is:

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + ... + P(A|E_n)P(E_n)$$

Proof.

$$\begin{aligned}
P(A) &= P(A \cap \Omega) \\
P(A) &= P(A \cap (E_1 \cup E_2 \cup ... \cup E_n)) \\
P(A) &= P((A \cap E_1) \cup (A \cap E_2) \cup ... \cup (A \cap E_n)) \\
P(A) &= P(A \cap E_1) + P(A \cap E_2) + ...P(A \cap E_n)
\end{aligned}$$

Law of Total Probability

For a set of mutually exclusive events, $E_1, \ E_2, \ E_3....E_n$, which together form the sample space Ω, the probability of any event A is:

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + ... + P(A|E_n)P(E_n)$$

Proof.

$$\begin{aligned}
P(A) &= P(A \cap \Omega) \\
P(A) &= P(A \cap (E_1 \cup E_2 \cup ... \cup E_n)) \\
P(A) &= P((A \cap E_1) \cup (A \cap E_2) \cup ... \cup (A \cap E_n)) \\
P(A) &= P(A \cap E_1) + P(A \cap E_2) + ...P(A \cap E_n) \\
P(A) &= P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + ... + P(A|E_n)P(E_n)
\end{aligned}$$

$\square$

Example: Law of Total Probability

Say you decide to implement a Democracy Promotion Program (DPP) in some villages in Myanmar in the hopes of increasing voter turnout at the next election. What is P(vote) after DPP?

Example: Law of Total Probability

Say you decide to implement a Democracy Promotion Program (DPP) in some villages in Myanmar in the hopes of increasing voter turnout at the next election. What is P(vote) after DPP?

- Let $P(vote|DPP) = 0.6$

Example: Law of Total Probability

Say you decide to implement a Democracy Promotion Program (DPP) in some villages in Myanmar in the hopes of increasing voter turnout at the next election. What is P(vote) after DPP?

- Let $P(vote|DPP) = 0.6$
- Let $P(vote| \sim DPP) = 0.3$

Example: Law of Total Probability

Say you decide to implement a Democracy Promotion Program (DPP) in some villages in Myanmar in the hopes of increasing voter turnout at the next election. What is P(vote) after DPP?

- Let $P(vote|DPP) = 0.6$
- Let $P(vote| \sim DPP) = 0.3$
- Let $P(DPP) = P(\sim DPP) = 0.5$ (Decided based on coin-flipping)

Example: Law of Total Probability

Say you decide to implement a Democracy Promotion Program (DPP) in some villages in Myanmar in the hopes of increasing voter turnout at the next election. What is P(vote) after DPP?

- Let $P(vote|DPP) = 0.6$
- Let $P(vote| \sim DPP) = 0.3$
- Let $P(DPP) = P(\sim DPP) = 0.5$ (Decided based on coin-flipping)
- $P(vote) = .6 * .5 + .3 * .5 = .45$

Example: Law of Total Probability

Infer $P(\text{vote})$ after mobilization campaign

Example: Law of Total Probability

Infer $P(\text{vote})$ after mobilization campaign

- $P(\text{vote}|\text{mobilized}) = 0.75$

Example: Law of Total Probability

Infer $P(\text{vote})$ after mobilization campaign

- $P(\text{vote}|\text{mobilized}) = 0.75$
- $P(\text{vote}|\text{not mobilized}) = 0.25$

Example: Law of Total Probability

Infer $P(\text{vote})$ after mobilization campaign

- $P(\text{vote}|\text{mobilized}) = 0.75$
- $P(\text{vote}|\text{not mobilized}) = 0.25$
- $P(\text{mobilized}) = 0.6; P(\text{not mobilized}) = 0.4$

Example: Law of Total Probability

Infer $P(\text{vote})$ after mobilization campaign

- $P(\text{vote}|\text{mobilized}) = 0.75$
- $P(\text{vote}|\text{not mobilized}) = 0.25$
- $P(\text{mobilized}) = 0.6; P(\text{not mobilized}) = 0.4$
- What is $P(\text{vote})$?

Example: Law of Total Probability

Infer $P(\text{vote})$ after mobilization campaign

- $P(\text{vote}|\text{mobilized}) = 0.75$
- $P(\text{vote}|\text{not mobilized}) = 0.25$
- $P(\text{mobilized}) = 0.6; P(\text{not mobilized}) = 0.4$
- What is $P(\text{vote})$?

Sample space (one person) =
{ (mobilized, vote), (mobilized, not vote), (not mobilized, vote) , (not mobilized, not vote) }
Mobilization partitions the space (mutually exclusive and exhaustive)

Example: Law of Total Probability

Infer $P(\text{vote})$ after mobilization campaign

- $P(\text{vote}|\text{mobilized}) = 0.75$
- $P(\text{vote}|\text{not mobilized}) = 0.25$
- $P(\text{mobilized}) = 0.6; P(\text{not mobilized}) = 0.4$
- What is $P(\text{vote})$?

Sample space (one person) =
{ (mobilized, vote), (mobilized, not vote), (not mobilized, vote) , (not mobilized, not vote) }
Mobilization partitions the space (mutually exclusive and exhaustive)
We can use the law of total probability

Example: Law of Total Probability

Infer $P(\text{vote})$ after mobilization campaign

- $P(\text{vote}|\text{mobilized}) = 0.75$
- $P(\text{vote}|\text{not mobilized}) = 0.25$
- $P(\text{mobilized}) = 0.6; P(\text{not mobilized}) = 0.4$
- What is $P(\text{vote})$?

Sample space (one person) =
{ (mobilized, vote), (mobilized, not vote), (not mobilized, vote) , (not mobilized, not vote) }
Mobilization partitions the space (mutually exclusive and exhaustive)
We can use the law of total probability

$$\begin{aligned} P(\text{vote}) &= P(\text{mob.}) \times P(\text{vote}|\text{mob.}) + P(\text{not mob}) \times P(\text{vote}|\text{not mob}) \\ &= 0.6 \times 0.75 + 0.4 \times 0.25 \\ &= 0.55 \end{aligned}$$

Example Law of Total Probability

Mixture Models: flexible modeling strategy.

Example Law of Total Probability

Mixture Models: flexible modeling strategy.

Two coins:

Example Law of Total Probability

Mixture Models: flexible modeling strategy.

Two coins:

- Fair: $P(H) = 1/2$

Example Law of Total Probability

Mixture Models: flexible modeling strategy.

Two coins:

- Fair: $P(H) = 1/2$
- Biased $P(H) = 3/4$

Example Law of Total Probability

Mixture Models: flexible modeling strategy.

Two coins:

- Fair: $P(H) = 1/2$
- Biased $P(H) = 3/4$

Draw a coin from urn ($P(\text{fair}) = 1/2$) and then flip.

$P(H)$?

Example Law of Total Probability

Mixture Models: flexible modeling strategy.

Two coins:

- Fair: $P(H) = 1/2$
- Biased $P(H) = 3/4$

Draw a coin from urn ($P(\text{fair}) = 1/2$) and then flip.

$P(H)$?

$\Omega = \{(\text{fair}, H), (\text{fair}, T), (\text{bias}, H), (\text{bias}, T)\}$

Example Law of Total Probability

Mixture Models: flexible modeling strategy.

Two coins:

- Fair: $P(H) = 1/2$
- Biased $P(H) = 3/4$

Draw a coin from urn ($P(\text{fair}) = 1/2$) and then flip.

$P(H)$?

$\Omega = \{(\text{fair}, H), (\text{fair}, T), (\text{bias}, H), (\text{bias}, T)\}$

$$P(H) \quad = \quad P(\text{fair}) \times P(H|\text{fair}) + P(\text{bias}) \times P(H|\text{bias})$$

Example Law of Total Probability

Mixture Models: flexible modeling strategy.
Two coins:

- Fair: $P(H) = 1/2$
- Biased $P(H) = 3/4$

Draw a coin from urn ($P(\text{fair}) = 1/2$) and then flip.
$P(H)$?
$\Omega = \{(\text{fair}, H), (\text{fair}, T), (\text{bias}, H), (\text{bias}, T)\}$

$$\begin{aligned} P(H) &= P(\text{fair}) \times P(H|\text{fair}) + P(\text{bias}) \times P(H|\text{bias}) \\ &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{3}{4} \\ &= \frac{5}{8} \end{aligned}$$

Example Law of Total Probability

Mixture Models: flexible modeling strategy.
Two coins:

- Fair: $P(H) = 1/2$
- Biased $P(H) = 3/4$

Draw a coin from urn ($P(\text{fair}) = 1/2$) and then flip.
$P(H)$?
$\Omega = \{(\text{fair}, H), (\text{fair}, T), (\text{bias}, H), (\text{bias}, T)\}$

$$\begin{aligned} P(H) &= P(\text{fair}) \times P(H|\text{fair}) + P(\text{bias}) \times P(H|\text{bias}) \\ &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{3}{4} \\ &= \frac{5}{8} \end{aligned}$$

Mixture of two coins

Bayes' Rule

- $P(B|A)$ may be easy to obtain
- $P(A|B)$ may be harder to determine
- Bayes' rule provides a method to move from $P(B|A)$ to $P(A|B)$.

Conditional Probability and Bayes' Rule

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(B \cap A) = P(B|A)P(A)$$

Conditional Probability and Bayes' Rule

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(B \cap A) = P(B|A)P(A)$$

Theorem

Bayes' Rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

General Statement of Bayes' Rule

Let the events $A_1 ... A_k$ form a partition of the space S such that $P(A_j) > 0$ for $j = 1...k$. And let B be an event such that $P(B) > 0$. Then for $i = 1...k$

Theorem

Bayes' Rule

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{\sum_{j=1}^{k} P(A_j)P(B|A_j)}$$

Bayes' Rule Example

Suppose that the US has captured lots of foreign Islamic militants in Afghanistan:

- 20% from Saudi Arabia (A)
- 30% from Yemen (Y)
- 45% from Pakistan (K)
- 5% from the USA (U)

How likely is a foreign fighter to be a convert?

Further suppose that we know something about the probability that the foreign fighter is a ‘convert’ to Islam **C**:

- $P(C|saudiArabia) = 0.03$
- $P(C|Yemen) = 0.05$
- $P(C|paK) = 0.07$
- $P(C|Usa) = 0.8$

How likely is a convert to be from the USA?

$$P(USA|C) = \frac{P(U) * P(C|U)}{P(C)}$$

$$= \frac{P(U) * P(C|U)}{P(U) * P(C|U) + P(A) * P(C|A) + P(Y) * P(C|Y) + P(K) * P(C|K)}$$

How likely is a convert to be from the USA?

$$P(USA|C) = \frac{P(U) * P(C|U)}{P(C)}$$

$$= \frac{P(U) * P(C|U)}{P(U) * P(C|U) + P(A) * P(C|A) + P(Y) * P(C|Y) + P(K) * P(C|K)}$$

$$= \frac{.05 * .8}{.05 * .8 + .2 * .03 + .3 * .05 + .45 * .07} = .43$$

Even though 80 percent of American fighters are converts, relative to 7 percent of Pakistani or Yemeni or Saudi fighters, it is still more likely that the fighter is not from the US than from the USA.

How likely is a convert to be from the USA?

$$P(USA|C) = \frac{P(U) * P(C|U)}{P(C)}$$

$$= \frac{P(U) * P(C|U)}{P(U) * P(C|U) + P(A) * P(C|A) + P(Y) * P(C|Y) + P(K) * P(C|K)}$$

$$= \frac{.05 * .8}{.05 * .8 + .2 * .03 + .3 * .05 + .45 * .07} = .43$$

Even though 80 percent of American fighters are converts, relative to 7 percent of Pakistani or Yemeni or Saudi fighters, it is still more likely that the fighter is not from the US than from the USA. This is because it is unlikely for there to be a fighter from the US in the first place.

Monty Hall Problem

Suppose we have three doors. A, B, C.

Monty Hall Problem

Suppose we have three doors. A, B, C.
Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats.

Monty Hall Problem

Suppose we have three doors. A, B, C.
Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats.

- A contestant guesses a door.

Monty Hall Problem

Suppose we have three doors. A, B, C.
Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats.

- A contestant guesses a door.
- The host opens a different door, revealing one of two goats. The contestant has option to switch

Monty Hall Problem

Suppose we have three doors. A, B, C.
Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats.

- A contestant guesses a door.
- The host opens a different door, revealing one of two goats. The contestant has option to switch
- Should the contestant switch?

Monty Hall Problem

Suppose we have three doors. A, B, C.
Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats.

- A contestant guesses a door.
- The host opens a different door, revealing one of two goats. The contestant has option to switch
- Should the contestant switch?

Contestant guesses A

Monty Hall Problem

Suppose we have three doors. A, B, C.
Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats.

- A contestant guesses a door.
- The host opens a different door, revealing one of two goats. The contestant has option to switch
- Should the contestant switch?

Contestant guesses A
$P(A) = 1/3 \rightsquigarrow$ chance of winning without switch

Monty Hall Problem

Suppose we have three doors. A, B, C.
Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats.

- A contestant guesses a door.
- The host opens a different door, revealing one of two goats. The contestant has option to switch
- Should the contestant switch?

Contestant guesses A
$P(A) = 1/3 \rightsquigarrow$ chance of winning without switch
If C is revealed to not have a car:

Monty Hall Problem

Monty Hall Problem

$$P(B|C \text{ revealed}) = \frac{P(B)P(C \text{ revealed}|B)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)}$$

Monty Hall Problem

$$
\begin{aligned}
P(B|C \text{ revealed}) &= \frac{P(B)P(C \text{ revealed}|B)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)} \\
&= \frac{1/3 \times 1}{1/3 + 1/3 \times 1/2} = \frac{1/3}{1/2} = \frac{2}{3}
\end{aligned}
$$

Monty Hall Problem

$$
\begin{aligned}
P(B|C \text{ revealed}) &= \frac{P(B)P(C \text{ revealed}|B)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)} \\
&= \frac{1/3 \times 1}{1/3 + 1/3 \times 1/2} = \frac{1/3}{1/2} = \frac{2}{3} \\
P(A|C \text{ revealed}) &= \frac{P(A)P(C \text{ revealed}|A)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)}
\end{aligned}
$$

Monty Hall Problem

$$
\begin{aligned}
P(B|C \text{ revealed}) &= \frac{P(B)P(C \text{ revealed}|B)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)} \\
&= \frac{1/3 \times 1}{1/3 + 1/3 \times 1/2} = \frac{1/3}{1/2} = \frac{2}{3} \\
P(A|C \text{ revealed}) &= \frac{P(A)P(C \text{ revealed}|A)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)} \\
&= \frac{1/3 \times 1/2}{1/3 + 1/3 \times 1/2} = \frac{1}{3}
\end{aligned}
$$

Monty Hall Problem

$$\begin{aligned}
P(B|C \text{ revealed}) &= \frac{P(B)P(C \text{ revealed}|B)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)} \\
&= \frac{1/3 \times 1}{1/3 + 1/3 \times 1/2} = \frac{1/3}{1/2} = \frac{2}{3} \\
P(A|C \text{ revealed}) &= \frac{P(A)P(C \text{ revealed}|A)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)} \\
&= \frac{1/3 \times 1/2}{1/3 + 1/3 \times 1/2} = \frac{1}{3}
\end{aligned}$$

You can double the chances of winning with switch

Joint Probability and Independence

- In statistics, we will often seek to infer facts about the world from the outcomes of repeated experiments.

Joint Probability and Independence

- In statistics, we will often seek to infer facts about the world from the outcomes of repeated experiments.
- To discover the systematic component of the experiment, we need to make assumptions about what affects what.

Joint Probability and Independence

- In statistics, we will often seek to infer facts about the world from the outcomes of repeated experiments.
- To discover the systematic component of the experiment, we need to make assumptions about what affects what.
- The most important and stringent of these is the notion of independence.

Independence

Definition

Suppose we have two events E_1, E_2*. We say these events are* *independent* *if:*

$$P(E_1 \cap E_2) = P(E_1)P(E_2)$$

Definition

Suppose we have a sequence of events $E_1, E_2, \ldots, E_n$*. We say the sequence of events is* *mutually independent* *if for each subset of the sequence,* $E_{i_1}, E_{i_2}, \ldots, E_{i_j}$

$$P(E_{i_1} \cap E_{i_2} \cap \ldots \cap E_{i_j}) = \prod_{m=1}^{j} P(E_{i_m})$$

For a sequence to be independent, every subset must be independent

Independence and Information

Does one event provide information about another event?

Independence and Information

Does one event provide information about another event?

Definition

If learning about B does not change the probability that A occurs, A is independent of B.

$$P(A|B) = P(A)$$

Independence and Information

Does one event provide information about another event?

Definition

If learning about B does not change the probability that A occurs, A is independent of B.

$$P(A|B) = P(A)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

Independence and Information

Does one event provide information about another event?

Definition

If learning about B does not change the probability that A occurs, A is independent of B.

$$P(A|B) = P(A)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

Independence is symmetric: if F is independent of E, then E is independent of F

Examples of Independence

- Whether a red shows on the roulette wheel is independent of whether a green showed 10 times previously.

Examples of Independence

- Whether a red shows on the roulette wheel is independent of whether a green showed 10 times previously.
- The stock portfolio's performance relative to the market average today is independent of the performance yesterday.

Examples of Independence

- Whether a red shows on the roulette wheel is independent of whether a green showed 10 times previously.
- The stock portfolio's performance relative to the market average today is independent of the performance yesterday.
- The extent of rainfall in Mexico in 1664 is independent of Spanish investments in local elites in 1620.

Independence

What if A is *Mutually Exclusive* of B?

Independence

What if A is *Mutually Exclusive* of B?

$$A \cap B = \emptyset$$

Independence

What if A is *Mutually Exclusive* of B?

$$A \cap B = \emptyset$$

$$P(A|B) = \frac{P(\emptyset)}{P(B)} = 0 \neq P(A)$$

Independence

What if A is *Mutually Exclusive* of B?

$$A \cap B = \emptyset$$

$$P(A|B) = \frac{P(\emptyset)}{P(B)} = 0 \neq P(A)$$

If two events occur with positive probability, then they cannot be mutually exclusive and independent.

Example Independence Relationship

Flip a fair coin twice.

$$E = \text{first flip heads}$$

$$F = \text{second flip heads}$$

Example Independence Relationship

Flip a fair coin twice.

$E =$ first flip heads

$F =$ second flip heads

$$P(E \cap F) \quad = \quad P(\{(H,H),(H,T)\} \cap \{(H,H),(T,H)\})$$

Example Independence Relationship

Flip a fair coin twice.

E = first flip heads

F = second flip heads

$$
\begin{aligned}
P(E \cap F) &= P(\{(H,H),(H,T)\} \cap \{(H,H),(T,H)\}) \\
&= P(\{(H,H)\}) \\
&= \frac{1}{4} \\
P(E) &= \frac{1}{2}
\end{aligned}
$$

Example Independence Relationship

Flip a fair coin twice.

E = first flip heads

F = second flip heads

$$
\begin{aligned}
P(E \cap F) &= P(\{(H,H),(H,T)\} \cap \{(H,H),(T,H)\}) \\
&= P(\{(H,H)\}) \\
&= \frac{1}{4} \\
P(E) &= \frac{1}{2} \\
P(F) &= \frac{1}{2}
\end{aligned}
$$

Example Independence Relationship

Flip a fair coin twice.

E = first flip heads

F = second flip heads

$$\begin{aligned}
P(E \cap F) &= P(\{(H, H), (H, T)\} \cap \{(H, H), (T, H)\}) \\
&= P(\{(H, H)\}) \\
&= \frac{1}{4} \\
P(E) &= \frac{1}{2} \\
P(F) &= \frac{1}{2} \\
P(E)P(F) &= \frac{1}{2}\frac{1}{2} = \frac{1}{4} = P(E \cap F)
\end{aligned}$$

Independence: No Information

Suppose E and F are independent. Then,

$$\begin{aligned} P(E|F) &= \frac{P(E \cap F)}{P(F)} \\ &= \frac{P(E)P(F)}{P(F)} \\ &= P(E) \end{aligned}$$

Conditioning on the event F does not modify the probability of E.
No information about E in F

Independence and Compliments

Proposition

Suppose A and B are independent events. Then the events A and B^c are also independent.

Independence and Compliments

Proposition

Suppose A and B are independent events. Then the events A and B^c are also independent.

Proof.

$$P(A \cap B^c) \quad = \quad P(A) - P(A \cap B) \quad \textbf{LTP}$$

Independence and Compliments

Proposition

Suppose A and B are independent events. Then the events A and B^c are also independent.

Proof.

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A \cap B) \quad \textbf{LTP} \\ &= P(A) - P(A)P(B) \quad \textbf{Ind. of A and B} \end{aligned}$$

Independence and Compliments

Proposition

Suppose A and B are independent events. Then the events A and B^c are also independent.

Proof.

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A \cap B) \quad \textbf{LTP} \\ &= P(A) - P(A)P(B) \quad \textbf{Ind. of A and B} \\ &= P(A)(1 - P(B)) \end{aligned}$$

Independence and Compliments

Proposition

Suppose A and B are independent events. Then the events A and B^c are also independent.

Proof.

$$
\begin{aligned}
P(A \cap B^c) &= P(A) - P(A \cap B) \quad \textbf{LTP} \\
&= P(A) - P(A)P(B) \quad \textbf{Ind. of A and B} \\
&= P(A)(1 - P(B)) \\
&= P(A)P(B^c)
\end{aligned}
$$

□

Example: Independence and Causal Inference

Selection and Observational Studies

- We often want to infer the effect of some treatment
  - Incumbency on vote return
  - Democracy on war
- Observational studies: observe what we see to make inference
- Problem: units select into treatment
  - Simple example: enroll in job training if I think it will help
  - $P(\text{job}|\text{training in study}) \neq P(\text{job}|\text{forced training})$
- Background characteristic: difference between treatment and control groups
- Experiments: make background characteristics and treatment status independent

Example of Independence and Causal inference.

Example: democracy promotion programs in countries where we think they are likely to succeed

Example of Independence and Causal inference.

Example: democracy promotion programs in countries where we think they are likely to succeedSuppose a researcher got to choose where to put a democracy promotion program, and finds the following:

	Democracy	No Democracy
DPP	10	50
no DPP	2	13

Example of Independence and Causal inference.

Example: democracy promotion programs in countries where we think they are likely to succeedSuppose a researcher got to choose where to put a democracy promotion program, and finds the following:

	Democracy	No Democracy
DPP	10	50
no DPP	2	13

$$P(Democracy|\ DPP) = \frac{(10/75)}{(60/75)} = 1/6 = .17$$

$$P(Democracy|no\ DPP) = \frac{(2/75)}{(15/75)} = 2/15 = .13$$

Example of Independence and Causal inference.

Now if we knew that the researcher conditioned on a particular observable outcome, we could control for that variable:

Example of Independence and Causal inference.

Now if we knew that the researcher conditioned on a particular observable outcome, we could control for that variable:

		Democracy	No Democracy
Prior Good Gov	DPP	2	18
	no DPP	1	9
Prior Bad Gov	DPP	8	32
	no DPP	1	4

Example of Independence and Causal inference.

Now if we knew that the researcher conditioned on a particular observable outcome, we could control for that variable:

		Democracy	No Democracy
Prior Good Gov	DPP	2	18
	no DPP	1	9
Prior Bad Gov	DPP	8	32
	no DPP	1	4

$$P(Democracy|DPP\&GG) = \frac{(2/75)}{(20/75)} = .1$$

$$P(Democracy|noDPP\&GG) = \frac{(1/75)}{(10/75)} = .1$$

Example of Independence and Causal inference.

Now if we knew that the researcher conditioned on a particular observable outcome, we could control for that variable:

		Democracy	No Democracy
Prior Good Gov	DPP	2	18
	no DPP	1	9
Prior Bad Gov	DPP	8	32
	no DPP	1	4

$$P(Democracy|DPP\&GG) = \frac{(2/75)}{(20/75)} = .1$$

$$P(Democracy|noDPP\&GG) = \frac{(1/75)}{(10/75)} = .1$$

$$P(Democracy|DPP\&BG) = \frac{(8/75)}{(40/75)} = .2$$

$$P(Democracy|noDPP\&BG) = \frac{(1/75)}{(5/75)} = .2$$

Conditional Independence

Definition

Let E_1 and E_2 be two events. We will say that the events are conditionally independent given E_3 if

$$P(E_1 \cap E_2 | E_3) = P(E_1 | E_3) P(E_2 | E_3)$$

Proposition

Suppose E_1 *and* E_2 *and* E_3 *are events such that* $P(E_1 \cap E_2) > 0$ *and* $P(E_2 \cap E_3) > 0$. *Then* E_1 *and* E_2 *are conditionally independent given* E_3 **if and only if** $P(E_1 | E_2 \cap E_3) = P(E_1 | E_3)$.

Proposition

Suppose E_1 and E_2 and E_3 are events such that $P(E_1 \cap E_2) > 0$ and $P(E_2 \cap E_3) > 0$. Then E_1 and E_2 are conditionally independent given E_3 **if and only if** $P(E_1|E_2 \cap E_3) = P(E_1|E_3)$.

Proof.

Suppose E_1 and E_2 are conditionally independent given E_3. Then

$$\begin{aligned} P(E_1 \cap E_2|E_3) &= \frac{P(E_1 \cap E_2 \cap E_3)}{P(E_3)} \quad \textbf{Def of Conditional Pr.} \\ &= \frac{P(E_3)P(E_2|E_3)P(E_1|E_2 \cap E_3)}{P(E_3)} \quad \textbf{Mult. Rule} \end{aligned}$$

Proposition

Suppose E_1 and E_2 and E_3 are events such that $P(E_1 \cap E_2) > 0$ and $P(E_2 \cap E_3) > 0$. Then E_1 and E_2 are conditionally independent given E_3 **if and only if** $P(E_1|E_2 \cap E_3) = P(E_1|E_3)$.

Proof.

Suppose E_1 and E_2 are conditionally independent given E_3. Then

$$\begin{aligned}
P(E_1 \cap E_2|E_3) &= \frac{P(E_1 \cap E_2 \cap E_3)}{P(E_3)} \quad \textbf{Def of Conditional Pr.} \\
&= \frac{P(E_3)P(E_2|E_3)P(E_1|E_2 \cap E_3)}{P(E_3)} \quad \textbf{Mult. Rule} \\
P(E_1|E_3)P(E_2|E_3) &= P(E_2|E_3)P(E_1|E_2 \cap E_3) \\
P(E_1|E_3) &= P(E_1|E_2 \cap E_3)
\end{aligned}$$

□

Proof.

Suppose $P(E_1|E_2 \cap E_3) = P(E_1|E_3)$

Proof.

Suppose $P(E_1|E_2 \cap E_3) = P(E_1|E_3)$

$$P(E_1 \cap E_2|E_3) \quad = \quad P(E_2|E_3)P(E_1|E_2 \cap E_3) \quad \textbf{Multiplication Rule}$$

Proof.

Suppose $P(E_1|E_2 \cap E_3) = P(E_1|E_3)$

$$\begin{aligned} P(E_1 \cap E_2|E_3) &= P(E_2|E_3)P(E_1|E_2 \cap E_3) \quad \textbf{Multiplication Rule} \\ &= P(E_2|E_3)P(E_1|E_3) \end{aligned}$$

□

Going back to our democracy promotion example:

		Democracy	No Democracy
Prior Good Gov	DPP	2	18
	no DPP	1	9
Prior Bad Gov	DPP	8	32
	no DPP	1	4

Going back to our democracy promotion example:

		Democracy	No Democracy
Prior Good Gov	DPP	2	18
	no DPP	1	9
Prior Bad Gov	DPP	8	32
	no DPP	1	4

$$P(E_1|E_2 \cap E_3) = P(E_1|E_3)$$

$$P(Democracy|DPP\&GG) = (2/75)/(20/75) = .1$$

$$P(Democracy|GG) = (3/75)/(30/75) = .1$$